Camera QC Report

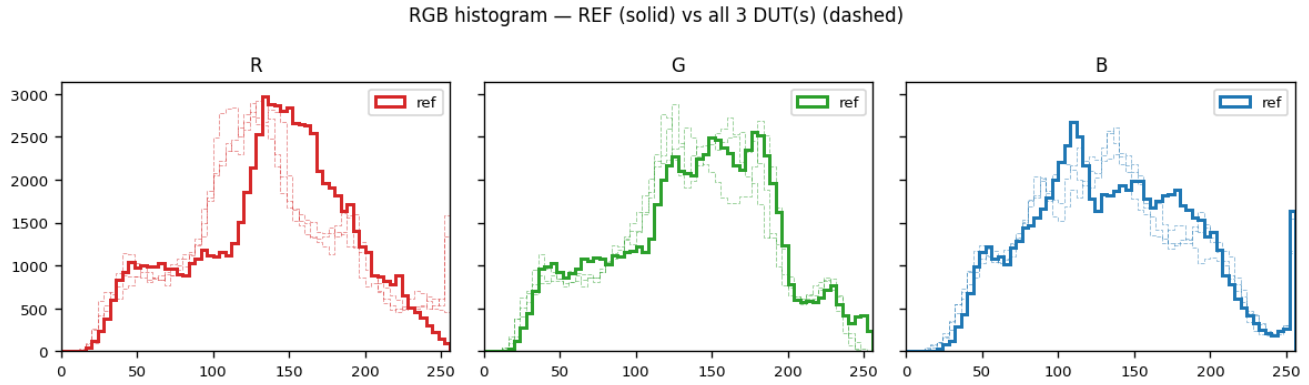
FAIL — DUT is worse than reference on multiple metrics.

- **Generated:** 2026-06-22 14:38 UTC
- **Scene preset:** `chessboard`
- **Reference:** `ref.jpg`
- **DUT images:** 3

Disclaimer — preliminary thresholds. The pass/marginal/fail thresholds used in this report are preliminary and should be tuned against a batch of known-good DUTs before this report is used as a standalone acceptance decision.

Summary (per metric, across all DUT images)

Metric	Direction	Tol %	Passed
L	closer to ref	10.0	3 / 3
a*	closer to ref	7.0	0 / 3
b*	closer to ref	12.0	1 / 3
R	closer to ref	10.0	3 / 3
G	closer to ref	10.0	3 / 3
B	closer to ref	7.0	3 / 3
sharpness	higher better	10.0	1 / 3
noise	lower better	0.0	3 / 3
sharpness_fft	higher better	10.0	1 / 3



Direction key: *higher better* — DUT failing means it dropped >tol% below ref. *lower better* — DUT failing means it rose >tol% above ref. *closer to ref* — DUT failing means |Δ| exceeds tol% of ref.

Per-DUT-image comparisons

DUT 1: `6000015-1221-012345-4040.jpg`

FAIL — two or more metrics outside tolerance. (4/9 metrics outside tolerance)

REF: ref.jpg



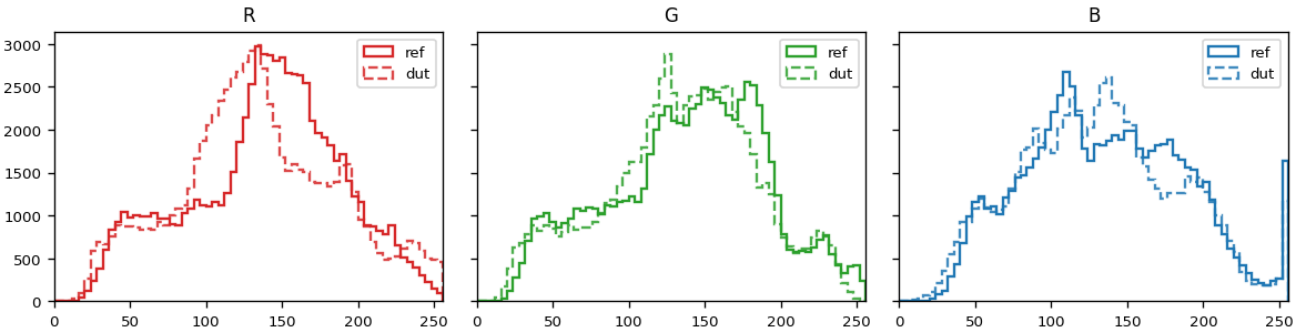
DUT: 6000015-1221-012345-4040.jpg



> **Conclusion:** sharpness 89.7 / 89.1% (avg ~89%); noise on par with ref; channels 3.0–3.7% darker; color stable.

Metric	Ref	DUT	Δ	% of ref	Tol %	Within tol?
L	138.02	133.28	-4.74	96.6	10.0	OK
a*	0.20	0.68	0.48	343.7	7.0	OUT
b*	1.79	1.53	-0.26	85.2	12.0	OUT
R	138.46	133.30	-5.16	96.3	10.0	OK
G	138.39	133.73	-4.66	96.6	10.0	OK
B	134.98	130.94	-4.05	97.0	7.0	OK
sharpness	654.38	586.98	-67.41	89.7	10.0	OUT
noise	8.35	7.98	-0.37	95.5	0.0	OK
sharpness_fft	2356.09	2099.87	-256.21	89.1	10.0	OUT

RGB histogram — REF (solid) vs DUT (dashed)



DUT 2: `6000037-1221-012345-4040.jpg`

MARGINAL — one metric outside tolerance. (1/9 metrics outside tolerance)

REF: ref.jpg



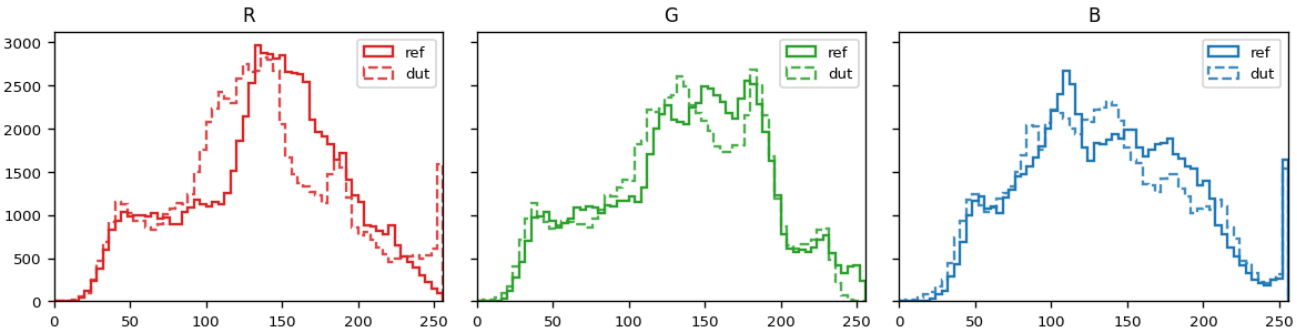
DUT: 6000037-1221-012345-4040.jpg



Conclusion: sharpness 94.1 / 93.2% (avg ~94%); noise on par with ref; channels 2.7–3.3% darker; color stable.

Metric	Ref	DUT	Δ	% of ref	Tol %	Within tol?
L	138.02	133.69	-4.33	96.9	10.0	OK
a*	0.20	0.59	0.40	300.2	7.0	OUT
b*	1.79	1.66	-0.14	92.3	12.0	OK
R	138.46	133.83	-4.63	96.7	10.0	OK
G	138.39	134.07	-4.32	96.9	10.0	OK
B	134.98	131.34	-3.64	97.3	7.0	OK
sharpness	654.38	615.75	-38.64	94.1	10.0	OK
noise	8.35	8.07	-0.29	96.6	0.0	OK
sharpness_fft	2356.09	2195.25	-160.83	93.2	10.0	OK

RGB histogram — REF (solid) vs DUT (dashed)



DUT 3: `6000099-1221-012345-4040.jpg`

FAIL — two or more metrics outside tolerance. (4/9 metrics outside tolerance)

REF: ref.jpg



DUT: 6000099-1221-012345-4040.jpg



Conclusion: sharpness 62.7 / 77.6% (avg ~70%); noise 82% of ref (cleaner); channels 1.2–2.4% darker; color stable.

Metric	Ref	DUT	Δ	% of ref	Tol %	Within tol?
L	138.02	135.30	-2.72	98.0	10.0	OK
a*	0.20	0.78	0.58	392.5	7.0	OUT
b*	1.79	1.40	-0.39	78.0	12.0	OUT
R	138.46	135.11	-3.35	97.6	10.0	OK
G	138.39	135.78	-2.61	98.1	10.0	OK
B	134.98	133.32	-1.66	98.8	7.0	OK
sharpness	654.38	410.11	-244.28	62.7	10.0	OUT
noise	8.35	6.82	-1.53	81.6	0.0	OK
sharpness_fft	2356.09	1829.35	-526.74	77.6	10.0	OUT

RGB histogram — REF (solid) vs DUT (dashed)

